ActInf Textbook Group ~ Cohort 1 ~ Meeting 15 (Chapter 6, part 1)

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All right, greetings everyone. It is September 9th, 2022. It's the 15th meeting for Cohort 1,
and we are discussing Chapter 6. We're in the first of two discussions on Chapter 6 of the textbook.
We're going to go to the questions page and start there, where we have one question that we explored
a bit last week in our welcome back, and then there's at least one question prepared, but anyone
is welcome to add more questions here while we turn first to this question, what are the four steps
in the recipe to construct an active inference model?
And can we make templates that facilitate people walking through some of these stages?
How much model building do people prefer to be engaged in in this section of the textbook group?
Do people want to have one or more model that they're personally developing?
Do they want those to be scaffolded in a shared page to help increment models along and see
models in different stages? Yeah, Brock, and anyone else?
It's the same problem as the first half of the textbook without math. It's not going to go well.
So this half without modeling, it's going to be quite difficult, I think. So I think we should just
get that out of the way that it's, yeah, it's not about whether you want to or not or whatever.
If you want to just understand it at a cursory, philosophical level, then maybe not. But if you
actually want to understand it, then yeah, we are going to have to model something some way. It
doesn't have to be the most complex, most precise one. But yeah.
I wonder if there are, if we could develop like maybe a couple examples or simple
ones that people might like the model.
I don't know.
Those are all going to be purely agent based, but
I think that's a good question.
Thanks.
Thanks.
I agree.
I wanted to frame it as a question, but I agree with that perspective.
And following along, as we structure a few specific cases, the rat and the T-maze, the eye
circadian, which are used in the textbook, then some of the examples that are used recurrently
in the literature, like birdsong and a few others.
And for these, many scripts exist.
And helping people who might not have the setup, either by with some walkthroughs.
Okay.
Here's how you get Octave running.
So you can do this MATLAB script, or here's the standalone DEM demo, because this textbook
really only gives the pointer to MATLAB scripts and methods.

Like when they say that the standard schemes can be applied, they mean a MATLAB script.

And that's exactly what the step-by-step guide, model stream one, is built around.

And chapter seven on modeling in discrete time is going to be akin to the step-by-step guide,

but it's not step-by-step-by-step-by-step.

It's more like it takes two bigger steps.

Okay.

How is this four-step recipe or active inference modeling similar or different than approaches

that people have seen for systems modeling from other frameworks?

For example, in a reinforcement learning or cybernetic modeling.

So something more recent and computational or something more pre-formal or computational.

Should this recipe be surprising to people coming from a certain background?

Are there sub-steps that are relevant to consider?

Brock, yes.

I don't have extensive formal modeling experience, but I just necessarily kind of most of the things

that I've done for the last half decade like entail some form of this.

And I don't see how you could model anything without going through these basic questions here.

Like this is, it's a bit, I don't see how this is specific or exclusive or whatever to active inference.

It's a bit ambiguous to me how that, you know, I mean, besides specifically the generative model, I guess.

But which system are we modeling?

Another way to ask that is what questions are we trying to answer?

Or who, I think Lyle brought this up.

Like who are you making the model for sort of thing?

Because the same, like you were saying just earlier about the particular choice of how you're modeling it,

it's just a choice on that system.

And that has, that entails, you know, particular kind of hidden states, Markov blanket, et cetera,

that may be relevant or may not be, depending on the questions, you know, what you're trying to model.

But that, that arises from those questions, I guess is what I'm saying.

Thank you.

Ali?

Well, yeah.

I just wanted to mention that, referring back to your previous question,

I also, I'm currently developing a model, actually an emotion perceiving agent by pretty much closely following those four steps.

But I'm getting some helps in that regard in how to actually implement those four steps in my particular situation.

But on the topic of modeling, well, at a very basic level, we can categorize different approaches to modeling in terms of answering to three different questions.

The what models, the how models, and the why models.

At least that's the kind of approach mostly done in neuroscientific modeling.

So, in my opinion, these, these four steps and this recipe can be applied mostly to the how and the why.

What models are mostly descriptive models of phenomena.

So, I think my connection, yeah.

What models are mostly, yeah, mostly descriptive models.

But when we want to model an agent, self-evidencing agent, which tries to,

which behaves as if it infers something about its environment, we have moved beyond just the descriptive stage.

And we're dealing with the how and probably even the why questions there.

Thanks.

When I hear why, I always think of Tinbergen's four questions, the four whys, Aristotle's four whys.

Why is a subjective modeler's pluralistic playground?

And it doesn't lend itself to one specific, even type of answer, let alone a framing or a specific parameter combination.

And I think that's a huge aspect of the move from, like, what are systems to how am I going to model the system,

which is a part of the turn that happens when we start to model, is we dispense with absolutism.

All of a sudden, these questions about what we're trying to do, our preferences and affordances and constraints and limitations and everything come into play.

And we might make a portfolio of models that are shining light on different aspects of even the very same phenomena.

So it's kind of like an empirically grounded pluralism and pragmatism when we start to be engaged in real useful modeling.

And that is something that I feel that one learns on the field, not necessarily bystanding.

So that's just to say that the modeling is really important, even for philosophical reasons.

To say nothing of some of the things that are coming to play in chapter 7 and beyond, where it's like,

yeah, the C vector has this many entries, because why?

And how many rows and columns does this have?

But once we understand why the rows and the columns and the shapes of these different entities are the way they are,

then we just have to use this scaffold to think about certain systems, consider their form at a very structural level,

like is it in discrete time or in continuous time and so on.

And then we might set up not just one generative model, but again, families of them.

In the future, we hope and expect that active blockference will facilitate us doing parameter sweeps across families of cognitive models and generative models.

But even in this boutique phase of model construction that we're in, in current year,

one would still say, okay, I want to have one model where it's either acceptable temperature or too hot.

And I want to have one with a three-state model.

Too cold, just right, too hot.

Here's going to be a four-stage model.

And just being able to specify that, again, instills pluralism through action about how we're modeling a system,

which is a philosophical point that it seems like sometimes isn't grasped by those who don't engage in the specific discussions around modeling a system.

Okay, so where does figure 6.1 come into play?

It's a rehearsal of some of the figures we've seen earlier in the book and many, many other places.

This is the particular partition.

The particle is the internal and the active sensory states.

So internal plus blanket states equals particular states.

Those are the particles.

If we were looking at dust under the microscope, like Brownian motion was describing initially,

the particles were the dust particle,

and then the external states were like that which was not the dust particle.

Partitioning the generative process and the generative model off from each other.

We're dealing in Bayesian mechanics with particular states, with particles, with things

that are able to engage in sensory motor type loops.

This is just bringing in a few more pieces than that, which is just to call attention to the direction and the existence of the arrows.

In the work, how particular is the free energy principle of Aguilera at all?

In the follow-up work, there's discussion around which topologies on this particular partition facilitate or enable which kinds of formal claims to be made.

For example, one might find it kind of interesting that sensory states have a bidirectional arrow with external states.

Or might find it equally interesting or differently interesting or less interesting that also active states have an arrow pointing back to internal states.

And, of course, the questions that we have raised all along the way of like,

okay, let's just say that we're going to model a person's arm.

What are the active states?

What are the sensory states?

What does it mean that there's a bidirectional arrow connecting them?

Is this the only topology for this particular partition?

Could we have the around-the-clock model?

External states influencing sensory, then internal, then active, and just going around the clock without any crosstalk in the blanket, without any back arrows.

Is that plausible?

Does that change the formalism?

On one hand, how could it not change the formalism?

When it's changing the structure of how variables are related to each other.

Or maybe there are formalisms that are actually abstracted away from any topology choice.

Because what is being described are the flows and the gradients on these variables.

But then what are those variables?

I think...

I mean, that just highlights this need for modeling.

But if we just really briefly go back to that.

It's like...

It's like...

The fact that there's this bidirectional arrow between external states and sensory states...

Is...

A bit confusing.

In that...

Again, the language...

Like...

Well, if it is acting...

On the environment...

Isn't that an action state?

You know...

Or something like this.

In what way is it bidirectional?

So could you not draw that as two...

You know, one arrow going to the sensory state.

And one arrow from the sensory state to back to the environment.

And then say...

What functionally does that entail?

And then you're doing this particular selection.

Of how you're modeling.

Right?

But I think you have to do that...

To understand why that is bidirectional there.

Right?

In that particular framing of...

The generative process, generative model there.

Right?

Which is...

Still to me also...

Slightly nebulous.

But I think...

Yeah.

Just...

The bidirectionality of it.

Between the action and sense states...

That kind of makes sense.

Almost.

That you sense your actions.

And your actions are a response to your sense.

But that's not really what's being done there precisely either.

Right?

So...

Just explicating that is what...

Yeah.

We need to...

Separate and explicate what...

Yeah.

Another really important question is like...

This action perception loop is appealed to...

Early and often.

It's the first picture in the textbook.

Before the high road and the low road in 1.2 is 1.1.

We're talking about agents and their interfacing with the niche.

And we're going to end up calling how we model agents a generative model.

And we're going to end up calling the niche.

And the dynamics of a niche.

And how it's influenced as the generative process.

And we're going to call the interface between the generative model and generative process a markout blanket.

And it's going to inherit this technical definition from Bayes graphs.

And we're going to add these variables.

And we're going to connect them a certain way.

But we're talking about the feedback loop with the agent and the niche.

Oh.

So what is this then?

Are the observations the sensory states?

And this isn't just like a low-hanging fruit.

Like, well then why isn't it...

Like, you know, why couldn't we call them...

I mean, they are called as X and Y.

Interestingly, in the continuous time formulation.

And I get it that they're being distinguished here because this is the continuous time POMDP and the discrete time POMDP.

But where do we see active states here?

We see policies.

So are active states along the branch influencing the B matrix?

How the hidden states change through time?

So...

So...

There may be a very simple trail from this action perception loop framing to the discrete and continuous time POMDPs.

Yeah.

I think...

Just to me, the simple...

If you...

Instead of thinking of this as a circular thing where you're literally going from one state to the next state to the way you might do in a normal state machine.

If you're thinking of, no, the active...

When you have an active state that...

However, it is changing.

That is necessarily entailing change in the sensory external state and the internal states.

And vice versa here with the sensory states.

It's like that if you perform some action of some kind, there's some active change in that state.

Like you're necessarily changing the internal hidden state.

Right?

So like if you...

You know, grab a glass of water or something like this.

Like what you're modeling now has changed slightly.

Right?

And the sensory...

Your internal states.

And the sensory states are doing the same thing where, you know, for the glass of water or whatever was this, you know, object just sitting there doing nothing.

But now it's become a thing to drink out of.

And that's changed in some sense the kind of external states that are being fed into your sensory states.

Right?

Like it's changed the Markov blanket that you're kind of selecting for or relating to.

And I mean that...

Yeah.

The PMDP, I mean, I think that is visibly there, you know, the same sort of thing.

Where if you have this observation, you're necessarily creating a new internal or sense state that is going to change your policy selection.

So...

And that's going to change your sense states.

So...

Returning to the recipe, one question that I think will reduce our uncertainty about as we really do the modeling is...

How do we account for residuals and understand which modeling residuals?

Which is to say, variability in data that are not explained by the model.

Which and what amounts of residuals are acceptable?

So if we were doing a linear regression and the only data we had on hand was height and the only data that we wanted to predict were weight.

Then we could just say we've used all the data and whether the model fits with an R value of 0.1 or 0.9.

We know what the residual is of height on weight regression.

So if we look at the regression and here's a number that describes what fraction of the variability was described.

With this regression, R value of 1 being like it's all perfectly on a line.

R value of 0 being like it's just a total scatter plot.

We can say how much variability is explained.

And then we're done with the empirical data that we had.

So we can't explain any more variability.

When we're looking at empirical traces of behavior, to explain even a small fraction of behavior for some systems,

it might be the case that the generative models already must be quite complex.

In another scenario, it might be the case that a very simple model, like looks to the left are 90% of the time followed by looking to the right.

I mean, on average, isn't that true?

So then when we are evaluating models, whether in block prints, sweeping across models, or just qualitatively or in a boutique way,

are we looking to reduce the residual to 0?

So what traces of data are we looking to explain variability in?

And then if we're not going to be, with empirical traces of behavior, looking to reduce our unexplained variance,

what is going to be, with empirical traces of behavior,

what is going to be the criteria by which we do model selection on?

The AIC and the BIC, which are information criteria,
balance variability explained across families of models with their degree of parameterization.
Penalizing having more parameters, rewarding better explanations.
So seeking to find a balance with models that are on the kind of Pareto frontier of explanatory and simple.
That's great.
That's like one extra nuance on top of merely explaining more variability with a smaller residual.
It also adds in the penalty for having more parameters.
So that, in theory and in practice, stops the modeler from just,
okay, well now we're going to add in the temperature in this other part of the world,
because 1% of the variability just happened to be explained.
So now we're just developing these like totally spiraling,
looking for more and more data sources to explain a decreasing amount of variability that's left.
So it's good to pull back from that edge through the use of information criteria.
But at the core of the criterion is still the imperative to reduce the residual,
to have a model that fits.
Within a model, the way that we fine tune it,
so that it fits data well, but not overfitting.
And then across models, we can see an analogous process of wanting the model structure that fits without,
for example, overfitting or over-including parameters.
So I think one question, which might be added here, is like, what data do we have?
If we're doing a didactic model, and we just want to sketch it out,
and run a simulation, and just purely generate data,
that might be useful in certain settings.
But the question of what behavioral data we have is very non-trivial,
if we're going to be approaching this as an empirical analysis problem.
Brock?
Just another question there.
What observations must we make?
Or something, is the, related to the, what data do we have?
If you don't have the data that,
if there's not sufficient affordance information contained in the data to model the problem, then...
Do we have to model our action selection
at a second or third order?
Yeah.

Exactly.
To find, yeah, to realize the right epistemic value.
Okay.

Okay.

So that was figure 6-1.

Discrete and continuous.

Shallow, deep, and hierarchical.

So just to address this, which might come up, shallow versus deep is describing how iterated

the model is through time within one type of time.

So you could have the 1-hour shallow model, where hours are units.

Or you could have the 10 hours, where it goes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Or you could have 100, depth of 100 with a time click of 1.

Or one could have a hierarchical model, where there's 10 hours that make like a deca hour.

And then there's 10 clicks of the deca hour to reach a planning horizon of 100 time steps.

So deep is describing within one unit or counter of time, how iterated that counter is.

Depth is describing strict hierarchical nesting.

And so they can both be used separately or together to describe events over longer and longer time horizons.

Crucially, this is in the context of language processing.

The duration of the word transcends that of any phoneme.

And the sentence transcends that of any word in the sequence.

And if we assume paragraphs are made of multiple sentences, paragraphs always transcend and encompass sentences.

So anyone else want to just add a point?

I'm sure that you fellows are familiar with this distinction.

But just wanted to point it out since temporality in these models and doing model selection on different forms of temporality is going to be an essential piece of the puzzle.

Yeah.

And it also reminds me of the Shankarian analysis theory that you remember we talked about earlier.

And it's hierarchical.

I mean, it's temporal hierarchy that tries to describe the whole musical piece as a kind of hierarchical layers,

which I think maps perfectly onto this specific example of language parsing.

Because that also is a kind of thing that should ideally be modeled in a very hierarchical way.

But the caveat here is unlike language, which is at least in some languages are necessarily hierarchical.

Music is at least tonal music or even non-tonal music are not necessarily, they don't necessarily have this hierarchical structure.

So depending on the kind of music or the genre of music, we might have either a very shallow and non-hierarchical structure or a very deep one.

So that's something that I think, again, refers back to the kind of system we're trying to model.

But even that kind of system needs to be more, I mean, specified in a more granular level in order to have an effective modeling strategy there.

Thanks for bringing in that domain.

Like, it makes me think of the intro, the chorus, the bridge, the verse.

Perhaps these are only in certain genre of music, but analogously.

And then the measure transcends any beat in the measure with a special case of like a 1-1 time signature or something and so on.

And then although language is described in a strictly hierarchical sense, I'm wondering how syntax and grammar actually create kind of a strange loop.

For example, it's trivial to say that phonemes are nested within words and sentences and so on.

But what about the semantics of a sentence that contains commas and exceptions and then at the end the speaker says the previous sentence is not how you think it is?

And that is also causing kind of like a deep linguistic recall that can be trivially modeled as containing nested units.

But will a model that narrowly considers the semantics of words, then the semantics of clauses, how will it be able to address some of these questions?

Here, I think we can point to a really excellent attribute of the active inference modeling framework.

So, in this discussion where the LW paper was cited, we could just marvel at nested systems and draw blankets on blankets and so on.

They could be drawn around every organelle and every little lipid granule in the neurons or the brain regions.

And so, we discussed a little earlier like doing structural model selection on which one of those are relevant.

However, this does not mean we need to attempt to model the entire brain to develop meaningful, pragmatic simulations of a single level.

For example, if we wanted to focus on word processing, we could address some aspects without having to deal with phoneme processes.

This means we can treat inputs from parts of the brain drawing inference about phonemes as providing observations from the perspective of word processing areas.

So, one could say, I'm going to make a figure 4.3 where S is the true word.

And phonemes are being passed as observations to my word inference engine.

And someone can say, but aren't phonemes the result of inference?

And someone can say, great, make that module in blockference, make that module and pass me that data.

And then we can do a nested model and we can graft those models together.

But through the Markov blanket formalism, not even the whole Markov blanket bounding a specific particular entity,

but just the trivial Markov blanket that exists insulating any two unconnected nodes in a base graph,

we can use that broader, still technical concept of a Markov blanket and just say,

all right, phoneme observations come in.

And surely we could make a model where another type of information comes in

and the inference is on phoneme and that's what's passed out.

So, I think, again, to point to this advantage of active inference modeling,

it allows us to situate phenomena as complex multiscale nested systems

and also bite off what we can chew

and makes it an empirical question of how grafting together which different structures might be useful.
But we don't need to go turtles all the way down.
We can just say, even if it were turtles all the way down,
we're modeling the third through the fifth turtle.
And the fifth turtle gets a top-down prior
and the third turtle gets sensory input from one below it.
And that's the part of the stack
and this is the lateral width that we're modeling.
Another advantage of active inference,
which I'm not even saying this is a unique advantage,
is that this goal-directedness and corollary capacities
like counterfactuals on goals
and the way that counterfactuals on world states influence our goal selection,
nested goals and transient goals or conditional goals
can be resolved in an unfolding action perception loop.
That's something that, as this example of, like,
wanting to enter an apartment,
of being goal-driven over multiple timescales
and then being able to, again, zoom in.
So it's like, oh, grab the keys.
Okay, well, here's the key-grabbing module.
And someone says, well, the key-grabbing is actually accomplished
through this kind of grasping behavior.
Then that could be modeled.
Or you can just say,
we're subsuming finger-grasping kinetics
in the grasping module
because the phenomena that we're trying to explain
and the data we have is just about when things were grasped.
Or maybe the only observations we have
is when apartments were entered.
So we can actually, I expect,
frame systems extremely expansively
and clarify where we're doing formal modeling.
I think maybe this is just a philosophical
or ontological framing of it,
but I don't see how you can model anything well
without admitting that it is either, you know,

incomplete or partially consistent or, you know,
but there is some hidden state
which is probably not included in the model.
And to me, active inference kind of natively
has an affordance,
at least, at the very least, a bookmark to do that.
And so whatever depth or hierarchy you modeled the system,
it's okay.
It's okay that there's still hidden state
and it's incomplete.
If it sufficiently models,
answers the question at hand,
then you can always add another layer
or, you know, make the model more complex.
And that's just what we were going to do anyways.
Nice comment.
I totally agree with that.
Again, thinking back to the linear model example,
where is that extra information
about, like, adjacent possibles for the model?
It's unstructured.
It's kind of like either the structured information
is going to enter your linear regression just so
and everything that's outside of your linear regression
is we're going to need to go back to the square one.
If we include a second observable,
we're going to have to go to square one
and do the model selection all over again
and test for all the repeat interactions again.
But in these multi-scale models that we're discussing,
like you said it was a bookmark,
it's kind of a nice way to frame it,
which is like the phonemes.
You're bookmarking or leaving an open USB port or something.
Anything that outputs a phoneme will plug into that part of the model.
Anything that is able to hear
can be plugged in to the frog riveting.
Anything that can see can be plugged into the visual component of the frog.

And so it's almost like consistent with all of this blanket talk,
we're being clear about what is modeled formally
and the borders of the formal model
are degrees of freedom
as formulated to be expanded upon
and composed.
One other...
I'm not sure the context is something
or Ben Gertl say,
attributed to...
Peng Shui,
but Pei Wang,
that in circular reasoning,
when your circle gets big enough,
it becomes coherent.
Maybe another way of saying
that if your model is insufficient,
that if you
add enough states,
if you expand your blanket enough,
eventually,
if you're modeling something correctly,
I guess,
you know,
it can...
the general process
can match.
Your blanket states
can come into alignment.
Nice.
Some more active inference-isms.
So,
exterior-oceptive,
proprioceptive,
interoceptive,
we're going to use
active inference modeling.
Memory,

attention,
anticipation,
planning as inference,
we're going to use active inference.
that is what it looks like
to integrate
disparate fields
using one model.
One family of models.
So,
you know,
others can work out
some
fun ways to
communicate and frame it,
but I think
when people are like,
like,
how is this simplifying
when
it might feel like
it's bringing in
a lot
or even
stepping on
namespaces
that people already
have familiarity in
like we discussed earlier,
you could ask,
how is memory
related to attention?
Or,
perhaps more
saliently,
how will you model
how memory
is related to attention?

Are you going to go on
archive
and just look for
neural network architectures?
Are you going to read
a book written
before computers
were invented
and look at prose?
Are you going to
dive into the
psychoanalytic
tradition
or something
that's maybe
even clinically
biomedical?
What's the move?
And active inference
addresses that.
In these
theme of
words that people
use that mean
different things
for 400,
we have
Bayes
optimal behavior.
well then,
why do things
go wrong?
And there's
multiple layers
to unpack.
First is,
going wrong
is about

your
perspective
on how the
system,
perhaps how you
prefer it to be
or how you
expected it to be.
But,
like,
given where
you put
the
dish
when it
fell off
the table
that was
just
compatible
with gravity.
So,
it was
finding its
optimal
position
and
fragmentation
where you
placed it.
The ball
rolled downhill
as a consequence
of where it
was placed
and the
slope it
was on.

And
framing
behavior
in that
light,
we're
modeling
behavior
as rolling
to the
bottom
of a
bowl.
What's
the bowl?
What's
the ball?
Those are
the questions.
But,
it's like
chemical
reactions
proceeding
under Gibbs
free energy
minimization
when we
have
policy
selection
driven
by
variational
and
expected
free
energy

minimization.
But,
how can
things that
go wrong
be optimal?
Well,
it has to
do with
different
parameters
in the
model.
Given
how they
are,
the model
performs
optimally.
That doesn't
mean computationally
efficiently,
it doesn't
mean adequately,
it doesn't
mean it
satisfies,
but it
may be
framed as
base
optimal.
Fixed
and learned
behaviors.
This is
going to
come into

play in chapter 7. There's an amazing continuity in theory and in simple examples with perception and learning, which is to say that perception happens over faster timescales, or perceptive-like processes happen over faster timescales, while learning-like processes happen over slower timescales, or even just more generally, perception and learning refer to parametric update processes, and you might even be able to situate the same example both ways. Like if we see a ball move across our visual field, are we perceiving the location of the ball? Are we inferring the location of the ball? Or are we updating and learning our position of the ball? For those with a computational background, learning often equates to parameter updating. And so in that sense, any kind of dynamical perception is

learning.
So for
some
backgrounds,
the difference
between
perception and
learning will
be seamless.
For other
backgrounds,
those are
going to sound
like totally
different
processes.
I mean,
isn't
perception
when you
see the
book,
but then
learning is
like when
you understand
something about
the book.
And then
speaking from
a more
modeling
perspective here,
a fixed
model or
fixing a
parameter of
a model

reduces the
model's
complexity
immensely.
Or another
way to say
it, taking
a fixed
parameter and
making it
learnable
introduces
multiple
multiple
hyper
parameters.
And there's
no single
way to
address
parameter
learning.
You could
say, well,
we're doing
moving average
learning, like
a common
filter, or
just a simple
sliding window.
We're taking the
average of the
last three.
But now you have
to parameter
sweep across how
many.

data.
And with a
single trial
of data,
there may
be, it
may not be
clear which
learning strategy
is implemented.
Because especially
if we want to
differentiate learning
strategies empirically,
we would want to
see like multiple
comparable
trajectories
within and
amongst
individuals in
similar and
different contexts.
But now, you
know, our lab
has gotten
quite big.
And we're
doing quite
large model
selections.
even for
learning and
updating
parameters that
might just
conversationally
seem really

simple.
Like preferences
being updated.
So here's
this description
of when it
comes to
inference and
perception as
being fast
changes and
learning as
slower changes.
Though they're
being modeled in
a really
analogous way,
it is worth
noting in this
book we
exemplify rather
simple generative
models that are
defined using
tabular methods,
e.g. with
explicit matrices
or tensors for
priors and
likelihoods in
small state
spaces.
In comparison,
more sophisticated
GMs are being
developed in
machine learning,
deep learning,

robotics,
etc.
etc.
So is
active inference
deep learning?
Okay.
Good fellow,
generative adversarial
networks.
but with
Benji
and Mirza,
but
are we
saying those
are formally
active inference
systems?
Or are we
just saying
that we
could
think of it
like bird
song
conceptually?
and then
are we
just
adding a
wrapper
or a
descriptor
to the
GANs
that our
non-active

colleagues
are building?
Is it
shining any
new epistemic
or pragmatic
light on
GANs?
I mean,
here's a
whole paragraph
on GANs.
home is
behind.
Oh,
that's
chapter 10.
What happened?
But there's
still a whole
paragraph on
GANs.
Did you
see that?
how did
that happen?
Whoa.
I'm not
sure what
the optimal
way to
view a
PDF is
or that
one exists,
but I
don't think
it's a

browser.
It's not
bad,
but it's
not optimal.
Yeah.
Yeah,
I don't,
I would,
I'm interested
in the,
you know,
this,
again,
the generative
model.
Oh,
I know what
those are.
So,
generative
model,
like,
and how,
you know,
specifically
GANs here
and active
inference,
like,
is one,
is the
rectangle
a square,
sorry,
square,
rectangle,
rectangle,

not a
square here,
is that,
you know,
what we're,
is being
said,
or is
there a,
you know,
degreed,
a indexed
blanket here
of varying
degree?
Yep.
Yep.
As far
as I
know,
and also
you fellows
might know,
like,
this is being
pointed to,
variational
autoencoders
and variational
Bayesian
inference
are directly
implicated
as potentially
recursive
cortical
networks
are

and
world
models.
But,
if these
bridges
can be
made
solid,
like you
said,
with squares
and rectangles,
so that we
can say
all sum
or none
of
x
or y,
or here's
how you
project
this model
to that
model,
it's
going to
just,
like,
break a
dam
with
possibly
certain
applications
of
ActInf.

Any
final
comments
on
6?
And,
I think,
certainly
by two
weeks from
now,
let's
look at
the
SPM
implementation
and
look at
some of
those
that are
pointed
towards
in the
book,
as well
as some
of,
like,
the
epistemic
chaining
and
PyMDP
type
models,
as well
as anything

that anyone
else builds.
So,
any final
comments?
Yeah,
I just
wanted to
point out
the recent
work by
Fields
et al,
especially
one,
the two
papers
that
first and
was also
the co-author
of,
is,
one of
their main
objectives
is to
cross
that bridge
between
active
inference
and VAE
via the
CCCD
system.
So,
that bridge

they're talking
about in this
book is
already been
investigated
to be
filled.
So,
yeah,
they're pretty
interesting
developments going
in that regard.
forward.
Okay,
any last
notes before
I stop
the recording?
What is that
paper?
I'll send you
the link
from the
Discord,
I forgot.
I'll send you
notes.
Okay.
Maybe just
one,
I don't
know.
We haven't
tried to
make an
effort on
the code

to lay
down some
basic
examples from
the first
half,
but should
there be
some,
if we're
going to
model stuff,
should we
have a
subpage for
that or
put something
somewhere in
the code
for that?
I don't
know.
We've had
this code
section from
the beginning
and just
however it
should be
totally
modifiable,
full control.
So,
however this
and anything
subpages need
to be modified
and the

GitHub link
to a
textbook,
anything
that people
suggest,
just let
us know.
Anything
else?
Okay.